Blockchain and Federated Learning in P2P Energy Trading: Privacy Protection and Prosumer Incentives

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***Abstract*—Although the P2P power transactions using the multiagent deep deterministic policy gradient (MADDPG) algorithm has been extensively studied, there are still challenges in privacy protection and training incentives. Furthermore, the stability and efficiency of the strategy decreases when dealing with nonindependent identically distribution (Non-IID) data from heterogeneous prosumers. Therefore, this article proposes a blockchain-enabled asynchronous federated learning-MADDPG (BEAFL-MADDPG) framework designed to enhance the training efficiency of heterogeneous prosumers while safeguarding data privacy. The framework includes a novel P2P energy trading model that facilitates energy trading amidst incomplete information while ensuring privacy assurances. In addition, a BEAFL-MADDPG algorithm is proposed, which accelerates training processes and enables parallel computation among agents. This algorithm enhances the efficiency of algorithm and empowers the training of diverse prosumers. Furthermore, a blockchain-enabled training mechanism and prosumer incentive scheme are proposed that not only encourage prosumer engagement in training but also ensure traceable transactions without the need for trust among participants. These mechanisms promote transparency and integrity, fostering a collaborative and secure environment for energy trading. Simulation results demonstrate that the framework achieves peak load reduction through optimized P2P trading, maintains computation efficiency across discount rates, and ensures secure transactions via blockchain-based incentives. These practical benefits support scalable and sustainable community microgrid operations.**

***Index Terms*—Asynchronous federated learning (AFL), blockchain, P2P, privacy protection, prosumers.**

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NOMENCLATURE

Abbreviations

AFL	Asynchronous federated learning.
DNN	Deep neural network.
EMS	Energy management system.
MADDPG	Multiagent deep deterministic policy gradient.
MDP	Markov decision process.
Non-IID	Nonindependent identically distribution.
PV	Photovoltaic.
RES	Renewable energy source.

Indexes

i, j	Index of prosumer.
M	Set of prosumers whose generation is greater than demand.
N	Set of prosumers whose demand is greater than generation.
t	Index of time period.

Parameters

$\alpha_{i,t}^{mg}, \beta_{i,t}^{mg}, \gamma_{i,t}^{mg}$	Cost parameters.
Δt	Operating time period.
$\lambda_{grid,t}^{buy}$	Prices of electricity for buying in time period t.
$\lambda_{grid,t}^{sell}$	Prices of electricity for selling in time period t.
θ^{Q}	Network parameter.
$p_{i,t}^{PV,\max}$	Maximum values of PV output.
$p_{i,t}^{WT,\max}$	Maximum values of wind power output.
r_t	Estimating future rewards.
$r_{d,i}$	Downward ramp rates.
$r_{u,i}$	Upward ramp rates.

Variables

λ_t^{buy}	Electricity buying price of prosumer in the P2P market.
λ_t^{sell}	Electricity selling price of prosumer in the P2P market.
ϖ_t	Supply-demand ratio of prosumers in time t.
$p_{i,t-1}$	Generating power of prosumer i in time period $t-1$.

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$p_{i,t}$ Generating power of prosumer i in time period t.
$p_{i,t}^{PV}$ PV power output of prosumer.
$p_{i,t}^{WT}$ Wind power output of prosumer.

I. Introduction

WITH the advancement of distributed generation technologies, an increasing number of electricity users are transitioning from traditional consumers to prosumers capable of both producing and consuming electricity [1]. In this context, P2P energy trading mechanisms are introduced to facilitate energy exchanges among prosumers in community microgrid systems [2], [3], [4].

The clearning of P2P energy trading can be categorized into the following four main types:

1) traditional market models [5], [6];
2) blockchain-based trading mechanisms [7], [8], [9];
3) game theory approaches [10], [11], [12];
4) reinforcement learning methods [13], [14], [15].

Traditional market models are well established but lack adaptability. Blockchain mechanisms enhance transparency but come with high implementation costs. Game-theoretic approaches excel in strategic analysis but suffer from high model complexity. In contrast, reinforcement learning methods, particularly multiagent deep deterministic policy gradient (MADDPG) algorithms, demonstrate superior performance in dynamic and uncertain environments, which are capable of managing interactions among multiple prosumers and adapting to continuously evolving market demands [16]. Therefore, MADDPG algorithms are viewed as flexible and efficient solutions for P2P energy trading.

MADDPG algorithms can learn from historical data and real-time information to optimize decision strategies, achieving more efficient energy trading. Based on this, researchers have conducted numerous works on P2P trading and related fields. Sun et al. [17] proposed an improved MADDPG algorithm to achieve fair trading and privacy protection in P2P co-trading markets. The scheme improves the convergence of the trading process and optimizes the interaction between agents. Harrold et al. [18] proposed a P2P energy trading using the MADDPG in microgrid management, optimizing renewable energy use and reducing costs. The enhanced algorithm individualized reward functions based on game theory, enabling agents to develop unique policies for better performance in fluctuating demand and competition. Zhu et al. [19] proposed a novel MADDPG algorithm with integrated conditional value-at-risk constraints for Nash equilibrium estimation in microgrid P2P energy trading. Wang et al. [20] applied a MADDPG algorithm to manage bus voltages using static var compensators. Similarly, Zhang et al. [21] developed a multiagent static var compensator algorithm addressing autonomous voltage control, establishing heuristic-based agent partitioning to facilitate centralized training and decentralized execution for scalable data-driven learning. Kumar et al. [22] proposed a Lyapunov-based MADDPG method, which optimizes task distribution and radio resource allocation to minimize energy consumption and delay in vehicular edge computing environment. However, the aforementioned research faces the following three challenges as shown in Table I: 1) each electricity trading participant who provides a large amount of local data for model training is usually a distinct entity, raising concerns about data security; 2) participants in energy trading possess distributed decision-making capabilities and need to engage in collaborative efforts to achieve overall optimality; 3) nonindependent identically distribution (Non-IID) data of different participants may encounter varying data distributions and magnitudes, and the data imbalance in traditional MADDPG may lead to decreased model generalization performance.

While MADDPG excels in dynamic environments, its training framework faces three critical barriers in P2P energy trading: privacy vulnerability, heterogeneity inefficiency and coordination fragility. To address these challenges, federated learning (FL) is a promising solution [23]. FL enables model training on local devices, avoiding the risk of data transmission to centralized servers and safeguarding the privacy and security of data [24]. Furthermore, FL enables participants to learn locally, update models, and share knowledge post-global updates, thereby facilitating distributed decision-making and multiagent collaboration. In adition, FL can address different data distributions and imbalances in P2P energy trading. This enables models to better adapt to various data distribution scenarios and enhance overall efficiency and generalization performance. To accommodate the varying training capabilities and Non-IID data of heterogeneous participants in P2P energy trading, AFL is introduced [25]. In AFL, each participant can learn and update independently without waiting for all participants to complete a round of updates before synchronizing the global model. However, FL alone cannot guarantee trustworthiness in decentralized energy markets. Malicious prosumers may submit falsified local models during asynchronous updates, compromising global model integrity. The absence of tamper-proof audit trails also impedes traceability, while unreliable participants risk free-riding or data poisoning. Integrating blockchain with AFL directly mitigates these FL limitations: 1) its immutable ledger provides cryptographic audit trails for all local model updates, ensuring traceability and nonrepudiation; 2) consensus mechanisms enforce trustworthiness by requiring validation before model aggregation, preventing manipulation; 3) smart contracts automate incentive payouts only for verified contributions, disincentivizing free-riding. Blockchain technology empowers the federated learning system with decentralized control and management capabilities, eliminating centralized risks and enabling participants to collectively oversee the learning process. Crucially, unlike blockchain applications focused solely on secure data trading [9], our integration specifically targets the federated training workflow and prosumer incentives within the P2P energy market context.

In summary, this article proposes a blockchain-enabled asynchronous federated learning-MADDPG (BEAFL-MADDPG) which is a framework unifying AFL and blockchain to harden MADDPG against energy trading's unique constraints. Distinct from prior enhancements to MADDPG, such as those employing successor features for knowledge transfer [28], our approach fundamentally rearchitects the training process to leverage AFL and blockchain. This addresses the core challenges of heterogeneity, privacy, and incentives at the system level. First,

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TABLE I
COMPARISON OF DIFFERENT MADDPG ALGORITHMS APPLIED TO P2P TRANSACTIONS AND RELATED FIELDS

Scheme	Communication Mechanisms	Key Methods	Data Privacy	Non-IID Data Handling	Distributed Decision-Making
Ref. [16]	Localized asynchronous	Ratio of perception & learning	◐	○	◐
Refs. [17], [20]	Synchronous communication	D3PG and TD3	○	○	◐
Ref. [18]	Localized asynchronous	CVaR-constrained	◐	○	●
Refs. [19], [21]	Synchronous communication	Coordinator approximator	○	●	◐
Ours	Localized asynchronous	Blockchain & federated learning	●	●	●

○: Not Addressed. ◐: Partially Addressed. ●: Fully Addressed.

utilizing MDP for state modeling and decision-making enables prosumers to take optimal actions based on environmental conditions, thereby optimizing electricity trading strategies. Building upon this, AFL can train model locally and share only model parameters, without exchanging raw data. This practice safeguards prosumers' privacy and enhances data security. Asynchronous update mechanism reduces network latency and computational burdens. This enables participants to asynchronously update models according to their individual circumstances without the need for simultaneous training completion, thereby enhancing system flexibility. Finally, the mechanism without trust and consensus algorithms of blockchain, in conjunction with AFL's asynchronous features, enhance system security and overall efficiency. Encouraging prosumers to participate in mining, similar to contributing to the system, allows them to receive corresponding rewards, thereby motivating their active involvement in the model training process. Meanwhile, blockchain technology can record data and transactions during the training process to ensure transparency and traceability of training.

The contributions of this article are summarized as follows.

1) We propose a distributed FL-enabled P2P energy trading, which balances energy transactions among prosumers in an incomplete information environment while ensuring privacy protection.
2) We propose a novel AFL-based MADDPG algorithm that accelerates training by enabling agent-level parallel computation and asynchronous updates. Each agent operates independently in its local environment. This enables avoiding the need to wait for data from other agents and sharing only model parameters to enhance privacy.
3) We propose a blockchain-enabled AFL training mechanism that combines with a prosumer incentive mechanism. This approach integrates the training process with blockchain workflows, which enables encouraging prosumer participation, enhancing privacy protection, ensuring transaction traceability, and maintaining trust-free operations.

The rest of this article is organized as follows. Section II introduces the framework and modeling of P2P energy trading. Section III proposes the AFL-based DRL algorithm and its workflow on the blockchain. Section IV provides the experimental environment of the proposed scheme and the analysis of the effect of the algorithm. Finally, Section V concludes this article.

II. P2P TRADING FRAMEWORK

This section constructs a framework for P2P energy trading including a transaction model of prosumers, market constraints, and a trading mechanism based on game theory.

A. Modeling of Prosumers

The P2P trading framework consists of prosumers and P2P trading managers, who are the market maintainers. P2P energy trading helps the energy balance of the upper grid, which is the motivation for its maintenance. Each prosumer is installed with EMS, which is responsible for automatically accomplishing the bidding and load dispatching tasks of P2P energy trading. Each prosumer is categorized into buyers and sellers based on power shortage and power surplus, and both parties can offer prices in the P2P trading market, which is cleared by gaming. Prosumer can generate electricity from RES. Considering the accessibility of PV equipment and wind turbines, the share of RES will increase in the future. PV and wind power generation models of the prosumer i are defined as

$$0 \leqslant p_{i,t}^{PV} \leqslant p_{i,t}^{PV,\max}, \tag{1}$$

$$0 \leqslant p_{i,t}^{WT} \leqslant p_{i,t}^{WT,\max} \tag{2}$$

where $p_{i,t}^{PV}$ and $p_{i,t}^{WT}$ are the PV and wind power output of prosumer i in time t; $p_{i,t}^{PV,\max}$ and $p_{i,t}^{WT,\max}$ are the maximum values of PV and wind power output predicted based on historical data.

In addition, microturbines are used by prosumers to generate electricity. The cost function $C_{i,t}$ associated with electricity generation for prosumer i is defined as

$$C_{i,t} = \alpha_{i,t}^{mg} p_{i,t} + \beta_{i,t}^{mg} \frac{1}{2} p_{i,t}^2 + \gamma_{i,t}^{mg}, \tag{3}$$

$$p_{i,t}^{\min} \leqslant p_{i,t} \leqslant p_{i,t}^{\max} \tag{4}$$

where $p_{i,t}$ is the generating power of prosumer i in time period t; $\alpha_{i,t}^{mg}$, $\beta_{i,t}^{mg}$, and $\gamma_{i,t}^{mg}$ are cost parameters related to factors such as fuel, operation and maintenance. Subsequently, the marginal cost of prosumer i can be defined as

$$C_{i,t}^{mg} = \alpha_{i,t}^{mg} + \beta_{i,t}^{mg} p_{i,t}. \tag{5}$$

Therefore, the P2P offer of the prosumer without considering the grid tariff and supply-demand relationship can be defined as

$$\lambda_t^{supy} = \alpha_{i,t} + \beta_{i,t}^{mg} p_{i,t} \tag{6}$$

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where $\alpha_{i,t}$ is the intercept of the prosumer's generation cost, which serves as a decision variable and is assigned by prosumer i [26]. $\alpha_{i,t}$ takes values in the range of $[0, \zeta \cdot \alpha_{i,t}^{mg}]$. ζ is the decision variable interval scale factor.

The constraints to be satisfied by prosumer i in P2P trading are as

$$\begin{cases} p_{i,t-1} - p_{i,t} \leqslant r_{d,i} \cdot \Delta t \\ p_{i,t} - p_{i,t-1} \leqslant r_{u,i} \cdot \Delta t \end{cases} \tag{7}$$

where $r_{d,i}$ and $r_{u,i}$ are the downward and upward ramp rates, respectively; and Δt is the operating time period.

Prosumer i can satisfy their demand by purchasing electricity from the grid or from other prosumers j who have surplus electricity. In contrast, the surplus prosumers j can sell the electricity to others in a P2P manner to make profits. This trading behavior satisfies the following constraints:

$$p_{i,t}^{buy} = \sum_{j \in J} p_{i-j,t}^{buy} + p_{i,t}^{grid,in}, \tag{8}$$

$$p_{i,t}^{sell} = \sum_{i \in I} p_{j-i,t}^{buy} + p_{i,t}^{grid,out}. \tag{9}$$

The prosumer i satisfies the following power balance constraint

$$p_{i,t}^{buy} + p_{i,t} + p_{i,t}^{PV} + p_{i,t}^{WT} - p_{i,t}^{sell} = p_{i,t}^{L}. \tag{10}$$

P2P trading prices based on demand response can be used to incentivize prosumers to participate in P2P trading. The supply-demand ratio ϖ_t is the basis for measuring demand response and is defined as

$$\varpi_t = \frac{\sum_{i \in M} p_{i,t} + p_{i,t}^{PV} + p_{i,t}^{WT} - p_{i,t}^{L}}{\sum_{i \in N} p_{i,t}^{L} - p_{i,t} - p_{i,t}^{PV} - p_{i,t}^{WT}} \tag{11}$$

where ϖ_t is the supply-demand ratio of prosumers in time t; M is the set of prosumers whose generation is greater than demand; N is the set of prosumers whose demand is greater than generation.

The electricity buying price and selling price of prosumer in the P2P market are respectively defined as λ_t^{buy} and λ_t^{sell} as

$$\lambda_t^{sell} = \begin{cases} \frac{\lambda_{grid,t}^{buy} \lambda_t^{supy}}{(\lambda_{grid,t}^{sell} - \lambda_{grid,t}^{buy})\varpi_t + \lambda_{grid,t}^{buy}} & 0 \leqslant \varpi_t \leqslant 1 \\ \lambda_{grid,t}^{buy} & 1 < \varpi_t \end{cases}, \tag{12}$$

$$\lambda_t^{buy} = \begin{cases} \lambda_t^{sell} \varpi_t + \lambda_{grid,t}^{sell}(1 - \varpi_t) & 0 \leqslant \varpi_t \leqslant 1 \\ \lambda_{grid,t}^{buy} & 1 < \varpi_t \end{cases} \tag{13}$$

where $\lambda_{grid,t}^{sell}$ and $\lambda_{grid,t}^{buy}$ are the prices of electricity for selling and buying in time period t of the grid, respectively.

In addition, the price constraint that should be satisfied in order to ensure the rationality of trading in the P2P market is as

$$\lambda_{grid,t}^{buy} \leqslant \lambda_t^{buy} \leqslant \lambda_t^{sell} \leqslant \lambda_{grid,t}^{sell}. \tag{14}$$

These constraints dynamically shape agent decisions during MADDPG training: Power balance (10) and ramp limits (7) define feasible action spaces $\mathcal{A}_i$with violations penalized in rewards (17); supply-demand ratio ϖ_t (11) and grid price bounds (14) update the state s_t, enabling adaptive pricing; market clearing (15) guides the Nash equilibrium (18), where constraint satisfaction maximizes collective benefits.

Notice that when $\varpi_t = 0$, the supply and demand of electricity for prosumers in the P2P market is balanced. Therefore, prosumers only trade electricity with the grid. Their buying and selling price are $\lambda_{grid,t}^{sell}$ and $\lambda_{grid,t}^{buy}$, respectively; when $0 \leqslant \varpi_t \leqslant 1$, the supply of electricity in the P2P market is smaller than the demand, and electricity is traded among prosumers. The electricity selling price of prosumer is λ_t^{sell} that decreases as the supply-demand ratio ϖ_t increases. The electricity buying price of prosumer is λ_t^{buy} that decreases as the supply-demand ratio ϖ_t increases; when $\varpi_t > 1$, supply of electricity is greater than demand in the P2P market, and prosumers will sell excess electricity to the grid. In addition, to ensure the operation of P2P transactions, the buying price of electricity should be $\lambda_{grid,t}^{buy}$. Therefore, the P2P market clearing model under the condition that the cost of prosumer group is minimized is as follows:

$$\min \sum_{i \in I} \sum_{t \in T} C_{i,t} + \lambda_{grid,t}^{buy} p_{i,t}^{grid,in} - \lambda_{grid,t}^{sell} p_{i,t}^{grid,out}$$
$$\text{s.t. (1)–(14)} \quad . \tag{15}$$

B. P2P Trading Mechanisms and Prosumers Decision Making

The objective of prosumer participating in P2P trading is to maximize individual benefits. Therefore, a P2P trading market mechanism can be established through game theory. In this market, the trading time is divided into T time periods, and the duration of each time period is t. The specific trading includes the following three steps.

1) *Identification:* The prosumer i decides to buy or sell electricity during each time period based on its own electricity demand and generation. The game model of seller and buyer are formulated for different situations of prosumer, which are defined as

$$\mathcal{G}_t = \{\{1, 2, \ldots, I\}, \{p_{i,t}^{L}\}_{i \in I}, \{\lambda_t^{sell}\}_{i \in I}, \{\lambda_t^{buy}\}_{i \in I}, \{C_{i,t}\}_{i \in I}, \{R_i\}_{i \in I}\}, \tag{16}$$

where $\{1, 2, \ldots, I\}$ is the set of P2P electricity trading game prosumer subjects; $\{R_i\}$ is the set of benefits of prosumer.

2) *Information Reporting:* Prosumers upload the amount of electricity they expect to sell to reach a trading agreement with other prosumers. In addition, the prosumer needs to determine some trading conditions, e.g., trading time period t, initial trading price $\lambda_{t_0}^{sell}$, etc., to clarify the trading agreement.

3) *Final Decision:* Prosumers i solve the game optimization problem according to

$$\max_t \sum_{t \in T} (\lambda_t^{sell} p_{i,t}^{sell} - \lambda_t^{buy} p_{i,t}^{buy} - C_{i,t}). \tag{17}$$

Subsequently, the solved P2P trading plan is reported to the platform, so that the information among prosumers can be exchanged. As the P2P trading game progresses,

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the benefits of each prosumer are maximized, i.e., the Nash equilibrium state, which is defined as

$$R_i(\{p_{i,t}^L\}_{i\in I}, \{\lambda_t^{sell}\}_{i\in I}, \{\lambda_t^{buy}\}_{i\in I}) \geqslant R_i(\{p_{i,t-1}^L\}_{i\in I}, \{\lambda_{t-1}^{sell}\}_{i\in I}, \{\lambda_{t-1}^{buy}\}_{i\in I}). \quad (18)$$

Finally, λ_t^{sell} and λ_t^{buy} are updated and the EMS is published to the prosumers.

III. Proposed Blockchain-Based Asynchronous Federated Learning MADDPG Algorithm

In the process of P2P trading, each prosumer is heterogeneous with different decision-making capabilities and elements. However, they all have the same requirements of security, transparency and trustworthiness. Based on this, we propose a BEAFL-MADDPG algorithm in order to realize efficient, secure, transparent, and trustworthy P2P trading.

A. Multiagent Deep Deterministic Policy Gradient

In the P2P electricity market, each individual prosumer functions as an autonomous entity. The ultimate gain from transactions hinges upon the aggregated benefits accrued across various temporal intervals. The overall state of the prosumers is constantly changing over time and is influenced by the decisions made in the previous time period. Hence, employing the agent-based approach, the game process is characterized as a partially observable MDP (POMDP), subsequently addressed through DDPG algorithm. The reason for adopting the DDPG algorithm is that, as an actor-critic algorithm algorithm, it enables continuous decision making, which is the exact needs for P2P game transactions.

In each time interval t of the decision-making process within the game, each prosumer i observes a state $s_{i,t}$ that is determined by the P2P price of prosumer i versus its own demand, $s_{i,t} = (\lambda_{1,t-1}^{buy/sell}, \lambda_{2,t-1}^{buy/sell}, \ldots, \lambda_{I,t-1}^{buy/sell}, \sum p_{i,t}^L)$; the action variable $a_{i,t}$ that is determined by the decision variable of the cost of power generation; and receives a payoff $r_{i,t}$ is equal to the difference between the P2P market's revenue and the prosumer's generation cost.

The objective of each agent i is to maximize its cumulative payoff, defined as

$$\mathcal{P}_{i,t} = \sum_t^T \gamma^{t-1} r_{i,t} \quad (19)$$

where $\gamma \in [0, 1]$ is the discount factor that is used to measure the value of time and the present value of money. It represents the factor that discounts future cash flows to the current time, reflecting market uncertainty.

The DDPG algorithm constructs two approximation functions of the actor-critic algorithm based on deep neural networks (DNNs) to represent the policy function $\mu(s|\theta^\mu)$, i.e., actor network μ, and the action-value function $Q(s, a|\theta^Q)$, i.e., critic network Q, respectively. In addition, for the convenience of training, the action target network μ' with parameter $\theta^{\mu'}$ and the critic target network Q' with parameter $\theta^{Q'}$ are created as copies of networks μ and Q, respectively. The Bellman equation defines a recurrence relation for Q^μ as

$$Q^\mu(s_t, a_t) = \mathbb{E}\left[r_t + \gamma Q^\mu(s_{t+1}, \mu(s_{t+1}))\right]. \quad (20)$$

The gradient function for the update of parameter θ^μ in network μ is as

$$\nabla_{\theta^\mu} J = \mathbb{E}_{\mu'}\left[\nabla_a Q\left(s_t, a \mid \theta^Q\right)\big|_{s=s_t, a=\mu(s_t)} \nabla_{\theta^\mu}\mu\left(s \mid \theta^\mu\right)\Big|_{s=s_t}\right]. \quad (21)$$

In addition, the action-value network Q, is updated for minimizing the loss function, as follows:

$$L(\theta^Q) = \mathbb{E}_{\mu'}\left[\left(Q(s_t, a_t \mid \theta^Q) - y_t\right)^2\right] \quad (22)$$

and

$$y_t = r_t + \gamma Q(s_{t+1}, \mu(s_{t+1}) \mid \theta^Q). \quad (23)$$

In this article, MADDPG strategy gradient is introduced to solve the equilibrium problem of the prosumer trading game. Combined with the critical techniques of the DDPG algorithm mentioned above, the training workflow of MADDPG is as

1) *Initialization:* Each agent initializes the actor network μ for the current strategy and the critic network Q for estimating future rewards r_t, respectively. Each network has a corresponding target network μ' and Q' that are slowly updated with parameters θ^Q and θ^μ that are used to stabilize the training process.
2) *Observation of Environment:* Each agent observes the state of its own environment s_t and may consider the states and behaviors of other agents.
3) *Action Selection:* Each agent selects an action $a_t = \mu(s_t) + n_t$, where n_t is noise, based on its actor network μ and the currently observed state s_t.
4) *Environmental Feedback*: Environment gives the next state and reward based on the actions of all agents. Each agent receives its own reward r_t and new state information s_{t+1}.
5) *Experience Storage:* Each agent stores its own experience tuple (s_t, a_t, r_t, s_{t+1}) to its respective experience replay buffer.
6) *Learning:* The agent chooses $\mathcal{T}$ tuples in the buffer to compose a minibatch of experience data. This data contains s_j, a_j, r_j, and s_{j+1} that are used to train and update the agent's critic and actor networks. The method of using a mini-batch helps to improve learning efficiency, reduce variance during training, and utilize bulk data for more efficient gradient descent.
   Specifically, we calculate the target Q-value according to (23), which is the current reward plus the value of the next state, used as the target for the critic network. Then, we compute the loss function of the critic network according to (22) and minimize the loss function according to $\nabla_{\theta^Q} L$. The network parameters θ^Q are further updated according to $\theta^Q \leftarrow \theta^Q + l_r \nabla_{\theta^Q} L$, where l_r is the learning efficiency of the gradient descent. In updating

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the parameters of the target network, soft update is used, i.e., the parameters of the target network are smoothed to the parameters of the current network by a certain percentage as shown in (24). The update rate β is usually a very small value, e.g., 0.001

$$\theta^{Q'} \leftarrow \beta\theta^{Q} + (1-\beta)\theta^{Q'},$$
$$\theta^{\mu'} \leftarrow \beta\theta^{\mu} + (1-\beta)\theta^{Q'}. \tag{24}$$

At the end of the training, the decision variables for the prosumer's offer are as

$$\alpha_{i,t} = 0.5 \cdot \zeta(a_{i,t}+1) \cdot \alpha_i^{mg} \tag{25}$$

where $a_{i,t}$ is the output value of μ, which takes the value range of $[-1, 1]$ according to the definition of the DDPG algorithm; α_i^{mg} is the intercept of the marginal cost of prosumer i.

Critically, heterogeneity among agents (e.g., computational resources, local data distributions, and decision policies) is natively supported. Each agent i maintains independent actor μ_i and and critic Q_i networks, initialized and trained locally. This allows adaptations to agent-specific environments and Non-IID data. During federated aggregation (Section III-B), individualized weights ω_i (29) reconcile divergent learning progress, ensuring global model robustness against agent diversity.

B. Blockchain-Based Asynchronous Federated Learning

In order to solve the problems of P2P energy trading efficiency and privacy protection for heterogeneous prosumer. Blockchain technology is used in the proposed framework.

Blockchain fundamentally addresses three limitations of federated learning in P2P energy trading: 1) trust: Decentralized consensus and cryptographic immutability eliminate single points of failure, enabling verifiable model exchanges without third-party intermediaries; 2) transparency: All transactions (model updates, rewards) are recorded on-chain, providing auditable trails for global model integrity; 3) incentive Alignment: Dual token rewards (mining + FL participation) incentivize prosumers to contribute high-quality local models. Smart contracts automate reward distribution proportional to data utility, ensuring fairness.

Blockchain leverages tamper-proof smart contracts to autonomously execute critical workflows: First, they distribute incentives to prosumers based on participation frequency. Subsequently, these contracts enforce decentralized model validation by requiring cryptographic confirmation from $> 51\%$ of miner agents before accepting local updates. Furthermore, they coordinate global aggregation by triggering synchronization once validated models are recorded in a new block. These self-executing contracts eliminate intermediaries. For example, the reward module uses on-chain records x_i to disburse tokens, while the validation module rejects models failing consensus checks. This ensures trustless automation of incentives and data integrity. Therefore, its application in the field of electricity trading has great advantages.

The proposed BEAFL system consists of P2P electricity trading participants represented by N agents. In this system, each agent acts not only as a trainer, but also as a miner in the blockchain. First, each agent trains the local model and then publishes the model as a request transaction of the blockchain to the whole network. At the same time, the agent also acts as a miner in the blockchain system to mine the block containing all the local models that are ready to be aggregated. All agents start training and mining at the same time. Subsequently, the agent-validated models in the new block are recorded and cannot be tampered with. Finally, each agent performs a global aggregation in a decentralized environment with all shared models in the validated block to update its local model. In addition, to incentivize agents to participate more actively in FL for P2P trading, the blockchain provides two key rewards: 1) Mining Reward R_{mine} granted upon successful block validation; and 2) a Participation Reward R_{part} proportional to the agent's contribution frequency. These rewards are formally integrated into the agent's MDP reward function to incentivize desired behaviors. The blockchain incentive scheme comprises two reward components integrated into the agent's MDP: Mining Reward: A fixed reward granted for successfully mining a block containing aggregated models $R_{mine,i} = B \cdot 1_{mined,i}$, where B is a constant block reward, and $1_{mined,i}$ is an indicator function equal to 1 if agent i mined a block in the interval related to timestep t, else 0. Participation Reward: A reward proportional to the agent's cumulative participation count in global aggregation $R_{part,i} = k \cdot n_i$, where k is a positive reward coefficient, and n_i is the total number of times agent i has uploaded its local model to participate in global aggregation up to and including the participation leading to the reward. Integrated MDP Reward: The agent's immediate reward $r_{i,t}$ in the MDP is now defined as $r_{i,t} = r_{trade,i} + R_{mine,i} + R_{part,i}$, where $r_{trade,i}$ represents the original reward component based on energy trading profit (market revenue minus generation cost). State Space Enhancement: The participation count n_i is explicitly included within the observable state $s_{i,t}$ for each agent i.

The workflow of BEAFL is shown in Fig. 1.

1) *Local Training of Agent i:* Each agent i has a local dataset, i.e., state s_i. The goal of local training is to minimize the loss function L. Agents use their dataset s_i and the downloaded global model θ_{glb}^{μ} to iterate t times the DDPG to update their local model θ_i^{μ} for local training. If there are m transition samples participating in the mini-batch training, the formulas for calculating the strategy gradient and loss function are modified as

$$\nabla_{\theta^{\mu}} J = \frac{1}{m}\sum_{m} \nabla_a Q\left(s_{t=m}, a \mid \theta^{Q}\right) \nabla_{\theta^{\mu}} \mu\left(s_{t=m} \mid \theta^{\mu}\right) \tag{26}$$

and

$$L(\theta^{Q}) = \frac{1}{m}\sum_{m} \left(Q(s_{t=m}, a_{t=m} \mid \theta^{Q}) - y_{t=m}\right)^2. \tag{27}$$

Finally, the agent acts as a miner to publish the local model to the blockchain and mine at the same time. The participation count n_i for agent i is incremented upon publishing its local model.

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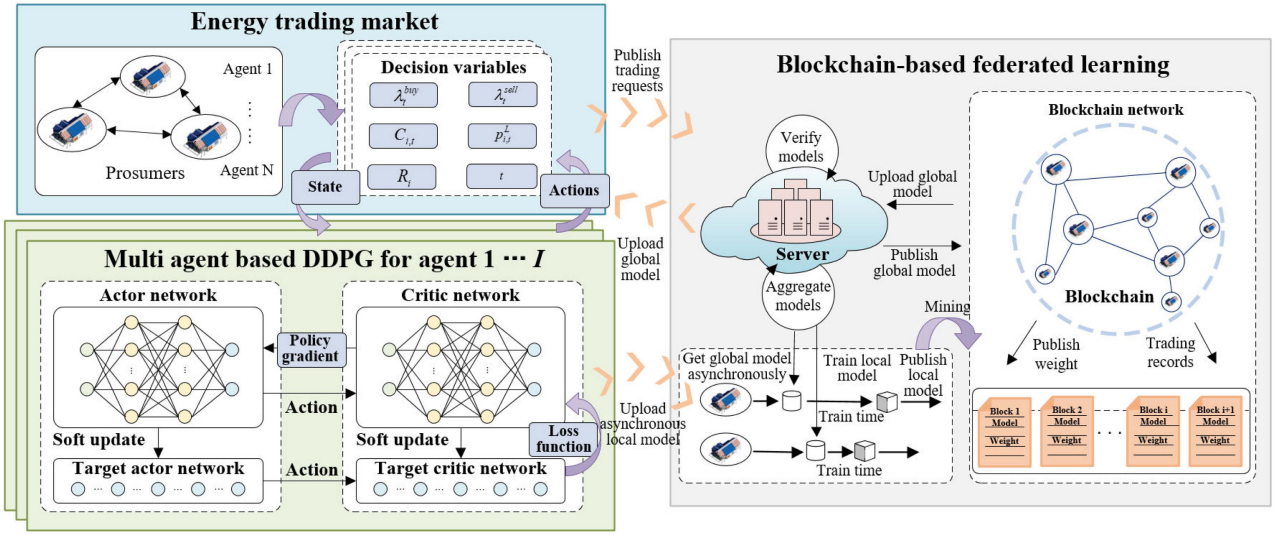


Fig. 1. Workflow of the BEAFL-MADDPG algorithm.

2) *Local Model Publishing and Validation:* Agent as a miner will write the local model to their new block and publish their local model while on the blockchain. When other intelligences acting as miners receive the model, they verify its accuracy and record it in their new block.
3) *The Agent's Mining Process:* All agents generate a global model after receiving models from other agents and then compete to mine new blocks.
4) *Global Model Updating:* In the AFL aggregation in this article, the number of local models to be aggregated in one training is limited to N. To prevent overfitting situations in the global model, the aggregation process is defined as

$$L_{glb}(\theta_{glb}^{Q}) = \sum_{i=1}^{N} \omega_i L_i(\theta_i^{Q}) \tag{28}$$

where L_{glb} and θ_{glb}^{Q} are the global loss function and global model parameters; L_i and θ_i^{Q} are the local function and global model parameters; ω_i is the aggregation weight of agent i, it is defined as

$$\omega_i = \frac{1}{x_i} \cdot e^{-\tau_i/\tau_{\max}} \Big/ \sum_{j=1}^{N} \frac{1}{x_j} \cdot e^{-\tau_j/\tau_{\max}} \tag{29}$$

where x_i is the number of times it participated in the training; $\tau_i = t_{\text{current}} - t_{\text{update}_i}$ is staleness duration, $\tau_{\max}$ is a tolerance threshold (set to two training rounds). This exponentially penalizes stale updates, preventing outdated parameters from skewing the global model. The design of ω_i is crucial for mitigating biases introduced by Non-IID data distributions across heterogeneous prosumers. Agents participating less frequently (lower x_i) typically possess more unique or underrepresented local data distributions, a common characteristic of Non-IID scenarios. Assigning a higher weight θ_i to these agents during global aggregation amplifies the contribution of their distinct local model updates θ_i^{Q} to the global model θ_{glb}^{Q}. This counteracts the potential dominance of models trained on more frequent but possibly less diverse data, enhancing the global model's ability to generalize across the entire heterogeneous prosumer population and improving its robustness against data distribution discrepancies.
5) *Blockchain Validation of Global Models:* All the agents compare the consistency of the received global models and the index of training participants. Agents will use the most results as the correct results, the rest of the uploaded messages are treated as malicious attacks and the intelligences that generated them are marked as malicious users for the record. Finally, all the agents record the correct global models and the index of the participating training intelligences into a new block.
6) *Block Generation:* The blockchain in this article uses proof of work (PoW) consensus mechanism. In this mechanism, blocks are generated by first collecting new transactions by nodes and packing them into a block. Next, the node tries to solve a mathematical problem that requires a lot of computational resources to solve, but the solution can be easily verified, a process called "mining." The node tries different random values as nonces, hashing them against the block's transaction data until it finds a hash that matches the conditions. The node then packages it with the block's transaction data and broadcasts it to other nodes in the network. Upon receiving the new block, the other nodes verify that its PoW is valid and that the transactions in it are legitimate. If the validation passes, the node adds the block to its own blockchain and continues the process of generating the next block.
7) *Global Model Download and Repeat:* Repeat the above process until the learning rate of the global model meets the demand or converge.

The flow of the BEAFL-MADDPG algorithm is described in Algorithm 1.

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Algorithm 1: Blockchain-Enabled Asynchronous Federated DDPG Algorithm for P2P Electricity Trading.

Input: Initial global model θ^{μ}_{glb}, staleness threshold $\tau_{\max}$, reward coefficient κ
Output: Optimized policy μ^*

for each agent $i \in I$ **do**
  Initialize actor μ_i, critic Q_i, target networks μ'_i, Q'_i
  Initialize participation count $x_i \leftarrow 0$, last update t_i^{update}
  for episode $n = 1$ to N **do**
    if new block $\mathcal{B}_k$ received **then**
      Extract validated models $\{\theta^{\mu}_j\}_{j\in\mathcal{J}}$ from $\mathcal{B}_k$
      Update $x_j \leftarrow x_j + 1, t_j^{\text{update}}$ for $j \in \mathcal{J}$
      Compute weights $\omega_j = \frac{1}{x_j} \cdot e^{-\tau_j/\tau_{\max}} / \sum(\cdot)$ ▷ ((29))
      Aggregate: $\theta^{\mu}_{glb} \leftarrow \sum_j \omega_j \theta^{\mu}_j$ ▷ ((28))
      $\theta^{\mu}_i \leftarrow \theta^{\mu}_{glb}$ ▷ Sync local model
    end if
    Reset environment, observe s_i
    for $t = t_0$ to $t_{\max}$ **do**
      Select action $a_t = \mu_i(s_t) + \mathcal{N}$
      Execute a_t, observe r_t, s_{t+1}
      Store transition (s_t, a_t, r_t, s_{t+1}) in replay buffer $\mathcal{D}_i$
      Sample minibatch $\mathcal{E} \sim \mathcal{D}_i$
      Update critic via $\nabla_{\theta^Q} L$ ((27)) and actor via $\nabla_{\theta^{\mu}} J$ ((26))
      Soft-update targets: $\theta^{\mu'} \leftarrow \beta\theta^{\mu} + (1-\beta)\theta^{\mu'}$
    end for
    Publish θ^{μ}_i to blockchain
    Attempt mining: Solve PoW for block $\mathcal{B}_{\text{new}}$ containing $\{\theta^{\mu}_i\}$
    if mined successfully **then**
      Broadcast $\mathcal{B}_{\text{new}}$, receive reward R_{mine}
      $x_i \leftarrow x_i + 1$ ▷ Update participation count
    end if
  end for
end for
return Optimized policy μ^*

IV. Simulation Analysis

This section aims to validate the effectiveness of the proposed BEAFL-MADDPG algorithm in managing P2P energy transactions. Additionally, it provides a performance comparison with three baseline algorithms and presents an ablative study to analyze the algorithm's components.

A. Experiment Environment and Setting

Computational setup: The experiments are run on a PC with Intel Core CPU i7-9750H @ 2.6 GHz and 16 GB RAM. We developed the proposed AFL system using the PyTorch 1.10.0 deep learning framework and set a blockchain with PoW consensus mechanism. Blockchain smart contracts are deployed on Remix IDE to test accuracy. The smart contracts are implemented on Ethereum using the Ganache framework and Geth. All prosumers in a P2P transaction are registered on the blockchain, and prosumers can only choose to buy or sell electricity during the transaction.

Baselines: We have three baselines: 1) the E-MADDPG algorithm [27] improves upon MADDPG by increasing the size of the experience pool with more learning experience and using high-priority experience replay, enhancing learning efficiency and effectiveness; 2) MADDPG-SFs [28] enhances MADDPG with successor features, using a shared GSFN to evaluate long-term features for task-specific value functions; 3) MADDPG-SFKT [28] improves exploration by transferring learned knowledge from source domains, optimizing joint policies efficiently in complex target tasks. It is a communication-efficient algorithm applied in federated learning environments that minimizes the expectation of discrepancies between clients and groups stochastically.

In comparison, the proposed BEAFL-MADDPG is superior in several ways. First, it ensures better data privacy and security through blockchain-enabled AFL. Second, in terms of training efficiency and system flexibility, our scheme benefits from parallel computing of agents. Finally, with the support of smart contracts, the transaction process is more convenient and transparent, and our incentive mechanism is more effective in encouraging consumer participation. The performance evaluation results show that the overall training efficiency and system throughput of BEAFL-MADDPG can meet the requirements of power scheduling, despite some instability in the early stage of training convergence when focusing on future returns. Therefore, with the functional advantages as shown in Table I, the performance is comparable to MADDPG-SFs, MADDDPG-SFKT, and E-MADDPG.

B. Effectiveness Study

We verifies the effectiveness of the established model for a community P2P trading system consisting of 6 prosumers. The IEEE-30 bus system is used as an example to analyze the case. The 6 prosumers are connected to the distribution network through buses 1, 2, 13, 22, 23, and 27, respectively, and the specific network parameters are given in [29]. The grid adopts a time-of-use (TOU) pricing strategy [30].

As shown in Fig. 2, the proposed scheme realizes P2P energy trading under the TOU pricing strategy. The prosumer's electricity consumption is higher during the peak daytime hours of 6–10 h and the peak nighttime hours of 17–21 h. At the same time, the price of electricity sold by the grid is higher under the TOU mechanism, and prosumer will take the form of P2P energy trading to satisfy its load out of the pursuit of profit maximization. In this case, the higher grid price causes the P2P internal trading price to rise, which promotes the enthusiasm of P2P trading, and the P2P trading power increases significantly. This effectively cuts the peak load of the grid.

The prosumer uses less electricity during the nighttime trough period of 0-5 h and the midday trough period of 11–14 h. At the same time, the grid sale price of electricity is lower. As a result, the P2P electricity transactions between prosumer become less.

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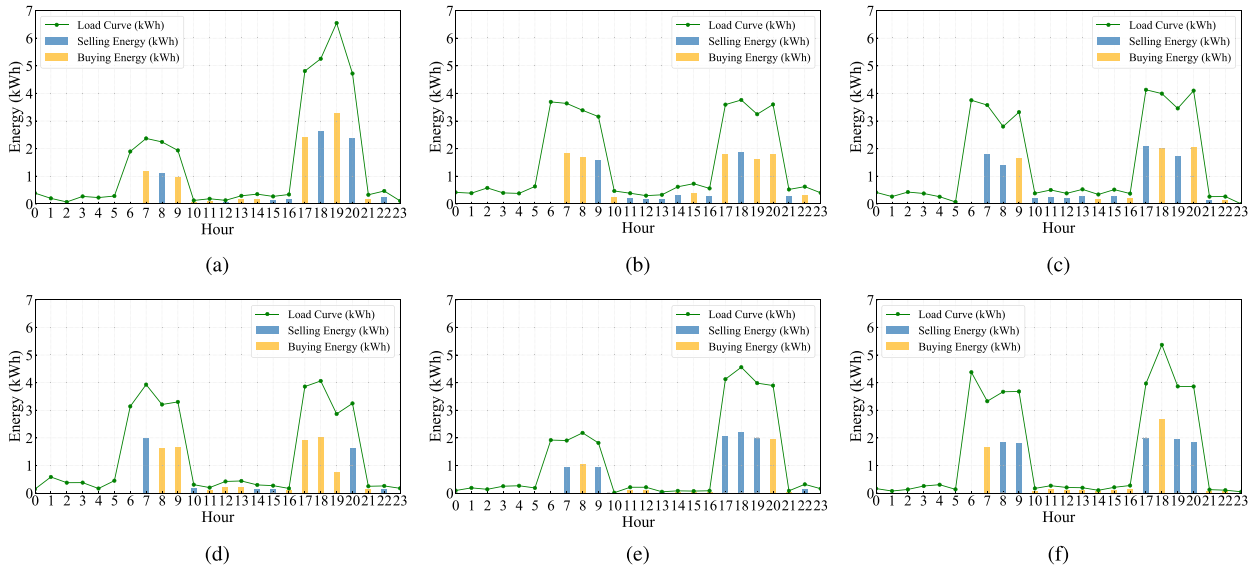


Fig. 2. Results of P2P energy trading for prosumers under the BEAFL framework. (a) Prosumer 1. (b) Prosumer 2. (c) Prosumer 3. (d) Prosumer 4. (e) Prosumer 5. (f) Prosumer 6.

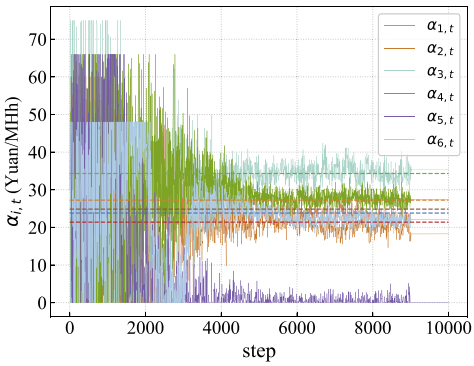


Fig. 3. Prosumers strategies for P2P trading Nash equilibrium.

TABLE II
P2P BENEFITS FOR PROSUMERS UNDER DIFFERENT OFFER STRATEGIES

Prosumer	$\alpha + 0.5$	$\alpha + 0.1$	$\alpha + 0.15$
Prosumer 1	21.6	27.5	29.3
Prosumer 2	34.4	56.1	50.1
Prosumer 3	41.4	33.1	43.9
Prosumer 4	41.5	33.4	42.9
Prosumer 5	14.6	21.2	18.6
Prosumer 6	20.8	19.7	23.9

In this case, the prosumer's load can be directly satisfied by the lower priced grid electricity.

When $\gamma = 0$, the prosumer's decision at each stage is considered independent of the other stages because it does not take into account the value of time. This makes the decisions at each stage similar to those in a static game. The decision making at each stage of the dynamic game in the above scenario can be considered similar to the decision making in the static game as shown in Fig. 3. The value of convergence is the Nash equilibrium of the static game. Thus, the converged value of the static game is used as a reference line that will appear in subsequent experiments with $\gamma \neq 0$ that take into account future uncertainty as a reference for the offer strategy.

Table II demonstrates the impact of the offer strategy action value α_i on the prosumer benefits. At a lower α_i, it means that the prosumer is willing to sell power at a lower price. This strategy is likely to be selected in the market clearing process, but the revenue may be lower. The low offer strategy will increase sales, but at a lower unit price, the total benefit will be lower. Boosting α_i will increase the prosumer's incentive to sell electricity, which means it expects to sell electricity at a higher price. During the market clearing process, this higher quoted prosumer may sell power at a higher price, resulting in a higher return, if demand and its prosumer's offer allow it. However, if the offer price is too high, it may result in its sale of electricity not being selected, and instead, it will not be profitable.

C. Performance Study

This section analyzes the performance of the proposed algorithm in terms of convergence effect and computation time. Fig. 4 gives a comparison of the convergence of different algorithms at $\gamma = 0.1$. It can be seen from the Figs. 4(a), (b), and (c) that the training of the other algorithms is affected due to the consideration of the uncertainty factor of the discount rate. Only the convergence of BEAFL-MADDPG remains good.

The training convergence of the proposed scheme with different discount rates γ is shown in Fig. 5. When γ changes from 0.5 to 0.9, the prosumer is more concerned about the future gain, which makes it more unstable and difficult to estimate. Due to the increase in the discount rate, the starting convergence process becomes more difficult and the final pricing strategies.

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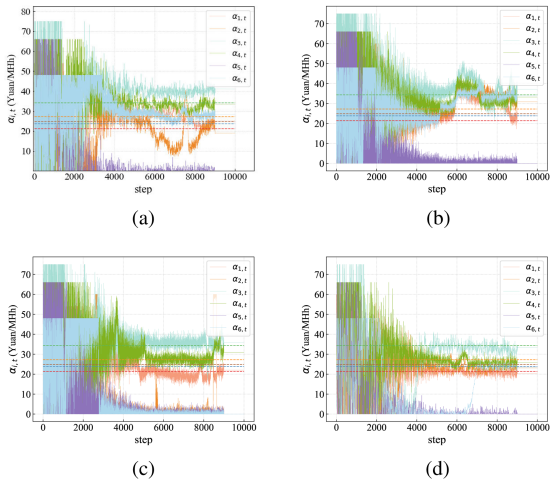


Fig. 4. Convergence curves of different MADDPG algorithms. (a) EMADDPG. (b) MADDPG-SFs. (c) MADDPG-SKFT. (d) BEAFL-MADDPG.

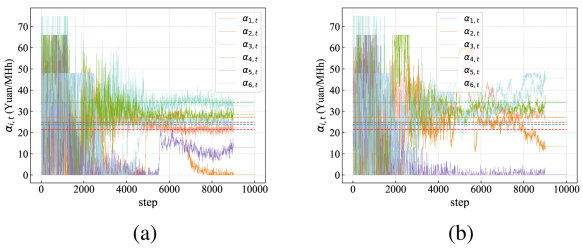


Fig. 5. Convergence curves of BEAFL-MADDPG with different discount rate. (a) $\gamma = 0.5$. (b) $\gamma = 0.9$.

TABLE III
COMPUTATION TIME

Discount rate	$\gamma = 0.1$	$\gamma = 0.5$	$\gamma = 0.9$
BEAFL-MADDPG	153.03s	151.58s	153.76s
EMADDPG	118.91s	140.5s	182.22s
MADDPG-SFs	121.61s	181.84s	234.19s
MADDPG-SKFT	191.61s	200.84s	204.19s

The change in the discount rate affect both the convergence and the total computational time. Table III reflects the change in computation time for different algorithms after increasing the discount rate, i.e., after accounting for uncertainty.

From Table III, it can be seen that the computation time of the proposed algorithm can satisfy the minimum time of the trading cycle of the power system i.e. 5 min. In addition, the algorithm adapts better to the uncertainty factor represented by the discount rate. Although, the proposed algorithm does not have an absolute advantage in terms of computation time in comparison with different algorithms, it shows a better adaptation with a shorter total computation time. To summarize, BEAFL-MADDPG not only has the functional advantages as mentioned in Table I in Section I, but also has the performance in terms of convergence effect, computation time and adaptability that is not inferior to the same type of algorithms in the baseline.

D. Ablation Study

To evaluate the individual contributions of each technical component of the proposed algorithm, we conducted a series of ablation studies. At the same time, we have taken into account the impact of the number of prosumers and communication delays on the system. Assuming that the communication delay of the blockchain system t_{delay} obeys $t_{\text{delay}} \sim \text{Exponential}(\lambda)$, $t_{\text{delay}} \in [0, t_{\text{delay}}^{\max}]$. Figs. 6 (a), (b), and (c) show the effect of the number of ablator-producers in the ablation experiments on the training time under different communication delays. The x-axis of the experimental plot represents multiples of the number of prosumer relative to the base system (i.e., 6 prosumers).

The combination of FL with MADDPG as shown by the red line in Fig. 6 can better utilize Non-IID, protect privacy and enhance the efficiency of multiintelligence collaborative learning compared to traditional MADDPG. Changing the communication method of federated learning to asynchronous as shown by the blue line in Fig. 6 improves flexibility, accelerates the learning process, reduces the communication delay, and enhances the fault tolerance of the system. Deploying the asynchronous federated learning-based MADDPG on the blockchain as shown by the green line in Fig. 6 can further ensure data privacy and security, enhance the degree of decentralization and transparency, motivate user participation, improve the efficiency of data utilization, and resist attacks, which significantly enhances the security of the system.

Since MADDPG is a method of learning through the interaction of an intelligent body with its environment, it usually involves decision-making and action selection at each time step. In the face of an increase in prosumer, communication delays can have a direct impact on the ability of an intelligent body to obtain feedback from the environment and make real-time decisions. The delay may result in the intelligences not being able to obtain timely updates on the state of the environment, thus affecting the accuracy and efficiency of their decisions.

FL allows model training on distributed devices or clients without the need to transfer data to a central server. In the face of an increase in prosumer, communication delays may lead to delays in data synchronization between different devices, affecting the speed and quality of global model convergence.

AFL in the face of client increase, the communication delay does not increase the time required for adversarial training between different clients. This is because each client can independently perform model updates during AFL training and then integrate the updated parameters through a centralized server. Therefore, the growth of prosumer does not have a direct impact on the overall learning process.

Blockchain-enabled AFL utilizes blockchain technology to ensure the security and transparency of model parameters and decisions. The same properties of AFL are available in the face of prosumer increase. However, communication delays may affect the speed of block generation and transaction confirmation time in the blockchain network, thus affecting the sharing and synchronization of model parameters. However, blockchain technology enhances data privacy and security, ensuring secure transmission of data and model parameters through distributed storage and encryption; secondly, the transparency

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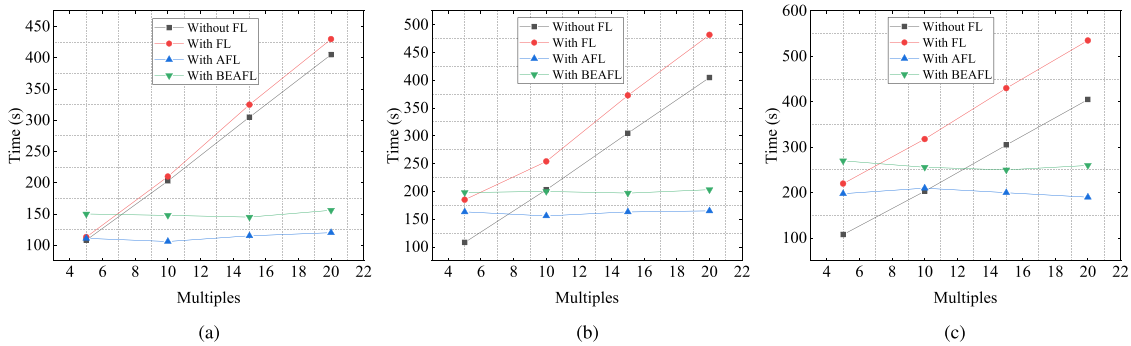


Fig. 6. Computational time ablation experiments with different communication delays and number of prosumers. (a) $\lambda = 5, t_{\text{delay}}^{\max} = 10$. (b) $\lambda = 7, t_{\text{delay}}^{\max} = 10$. (c) $\lambda = 150, t_{\text{delay}}^{\max} = 10$.

and traceability of blockchain make the whole learning process more credible and transparent; in addition, blockchain eliminates centralized control, which helps to improve the robustness and anti-attack ability of the system; and smart contracts regulate the protocols in the learning process and the reward and punishment mechanisms, facilitating participants to perform tasks according to rules.

Although the communication dilemma scenario increases the training time of the proposed scheme, the overall training time still meets the requirement of the shortest cycle time for power system scheduling under the number of prosumer demonstrated in the experiment which is the reasonable number of prosumer in the community.

V. Conclusion

This article addresses the challenges of privacy protection and training incentives in P2P power transactions using the MADDPG algorithm. The authors propose a BEAFL-MADDPG framework to enhance training efficiency for heterogeneous prosumers while safeguarding data privacy. The framework includes a P2P energy trading model that facilitates energy trading in an environment with incomplete information, ensuring privacy protection. In addition, an AFL algorithm is introduced to accelerate training processes by enabling parallel computation and asynchronous updates among agents, allowing each agent to operate independently in local environment and share only model parameters to enhance privacy. Furthermore, a blockchain-enabled training mechanism combined with a prosumer incentive scheme is proposed to encourage prosumer engagement and ensure transaction traceability. However, this work has limitations that warrant future investigation. First, the reliance on the PoW consensus mechanism introduces computational overhead and latency, which may constrain scalability in large-scale deployments. Second, the incentive model assumes rational participant behavior, whereas real-world scenarios may involve strategic manipulations. Finally, the framework's resilience under extreme network dynamics requires further validation.

In future work, enhancing scalability, robustness, and exploring advanced cryptographic techniques for stronger security and privacy are essential. In addition, integrating renewable energy forecasting models and demand response strategies could improve efficiency and grid stability.

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